



Techno economics and energy dynamics of a solar powered smart charging infrastructure for electric vehicles with advanced IoT based monitoring and RFID based security

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ARTICLE INFO

Keywords:

Charging station
Electrical vehicles
Energy economics
Energy management
Internet-of-things
Renewable energy resources
Scheduling, smart security
Sustainable development
Sustainable transportation

ABSTRACT

The widespread adoption of electrical vehicles (EVs) has led to an increased demand for charging stations which required an adequate charging infrastructure. However, this challenge brings substantial financial and ecological implications for the power grid and other major stakeholders such as EV users, system operators and policy-makers. To resolve this issue, this work proposes a smart EV charging infrastructure that provides 1) effective load management of EVs, 2) real-time monitoring of charging slots at a charging station, 3) smart security system, and 4) integration of renewable energy resources (RERs) into charging stations. This is achieved by setting out how to incorporate photovoltaic (PV) system into charging station infrastructure while suggesting effective EV load management with optimal energy utilization and minimal energy costs. Also, the annual power production of PV systems and power sharing between different power resources has been investigated to enable effective decisions about EV scheduling. The proposed infrastructure also offers an internet-of-things (IoT)-based monitoring of fast/slow charging units with real-time slot tracking to the users and charging station operators. The security aspect of the charging station is covered by a sensor-driven radio frequency identification (RFID) system. Experimental results that the proposed system generates 1,055,326 kWh/year, reducing grid reliance by 50% (to 335,775 kWh) and lowering CO₂ emissions from 425,326 kg to 212,210 kg. Moreover, energy costs drop from \$0.30 to \$0.13 per kWh and 270,567 kWh/year energy is sold back to the grid. The proposed system yields a net present revenue of \$1,020,000 which demonstrates significant economic and environmental benefits.

Introduction

EVs, charging infrastructures, and focus of this work

In the present era, the depletion of fossil fuel resources and environmental pollution arising from fossil fuel-based energy generation have become a major concern [1,2]. The transportation sector is heavily dependent on fossil fuels which pose numerous challenges concerning the availability of resources and environmental conditions [3,3]. Those

electrical vehicles (EVs) that are powered by batteries are proven as the optimal solution to minimize local urban air pollution by conventional fossil fuel-based transportation. However, the insufficient charging infrastructure for EVs presents a major hurdle in the widespread adoption of EVs [4,5]. This could bother the EV owners and can also affect the battery range of EVs. The large integration of EVs onto the power grid can increase carbon emissions which can lead to polluted gas emissions [6,7]. To address these challenges and minimize the polluted gas emissions renewable energy resources (RERs) are getting more and more attention. Nevertheless, the intermittent nature of photovoltaic (PV)

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<https://doi.org/10.1016/j.sfr.2024.100376>

Received 6 July 2024; Received in revised form 5 October 2024; Accepted 12 November 2024

Available online 14 November 2024

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Nomenclature

Acronyms

Alternating Current AC
 Battery Energy Storage System BESS
 Bayesian Neural Network BNN
 Comprehensive EV Charging Planning Model CCPM
 Cuckoo Optimization Algorithm COA
 Deep Maxout Network DMN
 Deep Reinforcement Learning DRL
 Deer Hunting Optimization and Crow Search Algorithm DHOCSA
 Demand-Side Management DSM
 Efficient Multi-Vehicle Scheduling Algorithm EMVSA
 Energy Management System EMS
 Embedded Systems Programming ESP
 Electric Vehicle EV
 Fractional Feedback Tree Algorithm FFTA
 Genetic Algorithm GA
 Greenhouse Gas GHG
 Grid-to-Vehicle G2V
 International Energy Agency IEA
 Internet of Things IoT
 Multi-deep Q-Network MDQN
 Particle Swarm Optimization PSO
 Photovoltaic PV
 Plug-in EV Charging Stations PEVCS
 Proportional Integral Derivative-Minimax Strategy with Hoo Control PID-MSH ∞
 Radio Frequency Identification RFID
 Renewable Energy Integration REI

Renewable Energy Resources RERs
 State-of-Charge SOC
 Time-of-Use ToU
 Vehicle-to-Grid V2G
 Vehicular Ad Hoc Network VANET
 Vanadium Redox Flow Battery VRFB
 World Health Organization WHO

Mathematical Symbols

T set of time slots representing the charging sessions,
 t time slot
 N set of EVs
 C set of charging stations
 P available power
 E energy demand
 B battery energy level
 Cost () cost function
 k set of EVs charging at the same station
 S_t availability of a charging slot
 M assignment of EV to charging slot
 A binary variable indicating authorized access
 U user presence at the charging station
 R power production from RERs
 S power-sharing between RERs, BESS, and the power grid
 G available power from the grid
 K cost savings achieved by utilizing RERs
 W indication variable if EV using RERs for charging

Chemical symbols

Carbon Dioxide CO₂

power generation introduces reliability concerns [8,9]. The focus of this work is the integration of RERs and advanced technologies into the charging infrastructure of EVs and to study the energy dynamics of the system.

CO₂ and technology switch towards EVs with zero carbon emissions

At the global scale, the primary generator of the polluted gas emissions is the transportation sector. It releases large amounts of carbon dioxide (CO₂) which is harmful to the environment [10,11]. The International Energy Agency (IEA) has reported that 23% of carbon emission has taken place in 2019 through the transport sector [12,13]. This huge amount of carbon emission highlights that it is a major concern that requires instant attention. An approach should be adopted to minimize the carbon emissions of the transportation sector. Within the transportation sector, the major source of carbon emissions are the cars and motor cycles and it emits 75% of the total carbon emissions [14,15]. Through the report, it is clear that to minimize the carbon emissions EVs should be adopted. Moreover, to charge these EVs renewable energy should be used that can minimize the carbon emission [16,17]. Similarly, the World Health Organization also mentioned that fossil fuel-based vehicles are the major contributor to carbon emissions that can affect people's health [18]. According to WHO 4.2 million deaths occurred in 2016 due to exposure to polluted air [19]. These facts and figures show that it is an urgent concern that needs to be addressed as soon as possible [20].

Moreover, EVs are getting attention worldwide and significant growth has been observed in recent years. In 2020, 43% increase was observed in the EVs growth which leads to the population of EVs to 10 million in the world [21]. The EV sales increase to 4.6% in China which increase the number of EVs to 4.5 million. Similarly, the EVs adoption also increased in the USA, Europe and Japan [22,23]. According to the

report, 420,000 EV buses were operational worldwide in 2020 and particularly in China. EVs offer environmental benefits such as zero carbon emissions and improved air quality. Moreover, with the integration of RERs, it contributes to a more sustainable transportation system [24].

Grid-tied solar-powered EV charging stations and motivation of this work

The integration of RERs into modern powered grids has been utilized as ancillary services. The integration has modes of distributed generation, EV batteries, PV systems, wind farms, etc. Among these renewable energy integrations (REIs), grid-tied solar-based EV charging stations are gaining widespread recognition due to their ability to produce cost-effective energy with minimal emissions [25]. The integration of grid-tied systems enhances their reliability and effectiveness which makes them capable of compensating the intermittent nature of solar power. In addition to these advantages, solar-based charging stations offer several benefits towards efficient scheduling, enhanced efficiency, and other EV charging-related activities [26]. The motivation for this work is a PV-powered charging infrastructure for EVs with adaptive features toward scheduling and security.

Literature review on charging stations monitoring and EVs scheduling

Numerous studies have been proposed in the literature for the efficient use of charging station monitoring and EV scheduling [27]. Examines using EVs as temporary energy storage systems, achieving a 90% reduction in grid energy consumption at Jordan University of Science and Technology through a proposed charging/discharging algorithm. The article [28] explores the financial benefits of using EVs as temporary energy storage systems in a microgrid with a photovoltaic plant. It demonstrates that G2V and V2G schemes can reduce grid dependency

and generate annual profits while maximizing the number of EV charging stations needed [29]. Introduces a multi-mode optimization technique for unidirectional EVs charging stations, designed to reduce charging costs in an open electricity market while integrating RERs across three operational modes: normal, eco, and boost [30]. Presents an innovative approach of integrating EVs into China's carbon markets while focusing on optimizing EV charging schedules based on carbon emission prices to enhance the economic feasibility of low-carbon transitions facilitated by EVs [31]. Examines the integration of EVs into the context of urban environments and the IoT for sustainable urban development. It proposes a model that utilizes historical data and deep learning methods to optimize energy management and increase net profit in the presence of EVs within the smart grid [32]. Proposes a hybrid genetic algorithm (GA) and particle swarm optimization (PSO) approach to optimize the placement of plug-in EV charging stations (PEVCS) in distribution networks with distributed generation. It aims to minimize power losses and voltage deviation. The simulation results validate the effectiveness of the GA-PSO approach while demonstrating the potential of EVs as a sustainable solution for reducing GHG emissions in the transportation sector [33]. Introduces a novel EV charging/discharging station (EVCDs) scheme utilizing clustered charging stations at commercial enterprise parking lots which effectively reduces the energy demand, high power demand, and the total energy cost for operators through incentivization [34]. Addresses the challenges of EVs towards optimal parking lot placement and forecasting EV performance in smart grids while utilizing a mixed approach with the Cuckoo optimization algorithm (COA) and Bayesian neural network (BNN) demonstrating effective solutions and accurate forecasting [35]. Offers a comprehensive review of the challenges and advancements in designing fast-charging stations for EVs emphasizing efficient charging speed, integration of renewable sources, and future expansion possibilities [36]. Addresses the problem of optimal bidding strategy selection for EV charging in an auction market proposing a multi-deep-Q-network (MDQN)-based reinforcement learning approach that achieves better economic benefits and reduced charging time compared to Q-learning and random approaches [37]. Provides a comprehensive review of EV battery charging methodologies, charging station levels, and advanced charging architectures, highlighting the integration of renewable energy and optimized energy management algorithms thereby proposing a future roadmap for EV adoption in India with research directions for power electronics-based solutions [38]. Develops a framework for optimal scheduling of an EV owner, considering car-sharing opportunities, telecommuting jobs, and charging cost minimization, and demonstrates its effectiveness in optimizing routing selections and charging time allocation with significant cost savings. The cost-optimal energy management model for campus microgrids while considering power generation and consumption analysis using IoT sensors. Its feasibility and effectiveness in reducing electricity cost and peak power through real-world datasets is proposed by [39].

The development and simulation of a bidirectional converter-based charging station for EVs, including vehicle-to-grid (V2G) and grid-to-vehicle (G2V) modes, cost calculation based on time of energy-based tariff, system monitoring of EV charging parameters, and implementation of an IoT-based system for data monitoring and retrieval are discussed in [40,41] presents a solar-battery storage integrated switchable glazing topology for the safe and efficient operation of EV charging stations in high-temperature environments. This includes solar PV integration, vanadium redox flow battery (VRFB) energy storage, internet of things (IoT)-based smart scheduling, and validation under various transient scenarios. An optimization-driven framework using the deep Maxout network (DMN) and the fractional feedback tree algorithm (FFTA) for EV charge scheduling in a vehicular ad hoc network (VANET) topology, considering factors such as predicted power, distance, average waiting time, and quantity of EVs charged has been proposed [42]. The proposed FFTA demonstrates improved performance in terms of distance, power, waiting time, and number of EVs charged

[43]. Proposes an optimal scheduling model with uncertainty consideration for EV charging load in coordination with the building air conditioning system and photovoltaic system. This eventually results in reduced non-guaranteed rates and load variance while maximizing flexibility and photovoltaic power consumption capacity. Similarly [44], introduces a comprehensive EV charging planning model (CCPM) and an efficient multi-vehicle scheduling algorithm (EMVSA) that addresses constraints of state-of-charge (SoC), driving direction, and enhances convergence speed using a novel reward shaping method in deep reinforcement learning (DRL)-based EV charging planning. A temporal-spatial scheduling model for EVs in AC/DC distribution networks is proposed in [52]. It considers the hybrid topology and constraints, aiming to optimize network losses and user inconveniences, achieving cost savings of 6.3% and 6.8% on two typical days. It also proposes a methodology for estimating time-of use (ToU) tariff rates using Big Data technology, enabling optimal demand side management (DSM) with EVs and achieving a 6% to 7% reduction in peak consumption [45]. Talks about the development of a data-driven optimization framework for integrating mobile chargers in urban transportation systems. It results in a 21% reduction in total costs and stabilization of the power profile through V2G implementation [46]. Covers the development and evaluation of a bi-directional V2G scheduling application that considers user preferences and grid operator demands to optimize EV charging/discharging. This results in improved load profiles, voltage profiles, distribution lines loading, and power losses in the low-voltage distribution network [47]. Utilizes an energy management system (EMS) for distribution systems using an IoT framework incorporating a hybrid technique called deer hunting optimization and crow search algorithm (DHOCSA) for optimal control of power and distribution system resources [48]. Proposes an advanced IoT-based DSM and control strategy for efficient energy management in smart microgrids. It integrates GA optimization with ToU pricing and proportional-integral-derivative (PID)-MSH ∞ control for reduced costs, improved regulation, and adherence to standards.

Despite the progress in EV charging infrastructure, several research gaps remain unaddressed in previous studies. First, many works do not integrate RERs into EV charging stations, limiting their ability to reduce grid reliance and carbon emissions. Second, while some studies focus on scheduling optimization, they lack real-time monitoring systems that provide up-to-date information on charging availability and usage. Third, the implementation of advanced security systems, such as RFID-based solutions is rarely considered. Fourth, existing research often overlooks cost-effective energy management solutions that use RERs to minimize operational costs. Lastly, there is a gap in exploring the use of automation to reduce labor costs, which is crucial for enhancing the economic feasibility of large-scale EV charging infrastructure.

Main contribution and graphical abstract of this work

Despite the substantial advancements achieved in the domain of EV charging infrastructure and scheduling optimization, there are still research gaps in the existing literature. These gaps include smart charging in the infrastructure for EV charging which could provide effective EV scheduling, real-time monitoring of the charging slots at a charging station, a smart security system in charging stations, and the integration of RERs into charging stations. To the best of the author's knowledge, such an integrated infrastructure of smart charging has not been proposed before in the literature. The main contributions of this paper are as follows:

- *Integration of Advanced Technologies:* The proposed PV-power smart charging station (PVPSCS) integrates the advanced technologies that revolutionize the EV charging infrastructure. The charging station minimizes dependence on traditional fossil fuels which results in a reduction of air pollution. The grid-tied feature of the system further enhances the reliability and effectiveness of charging stations that

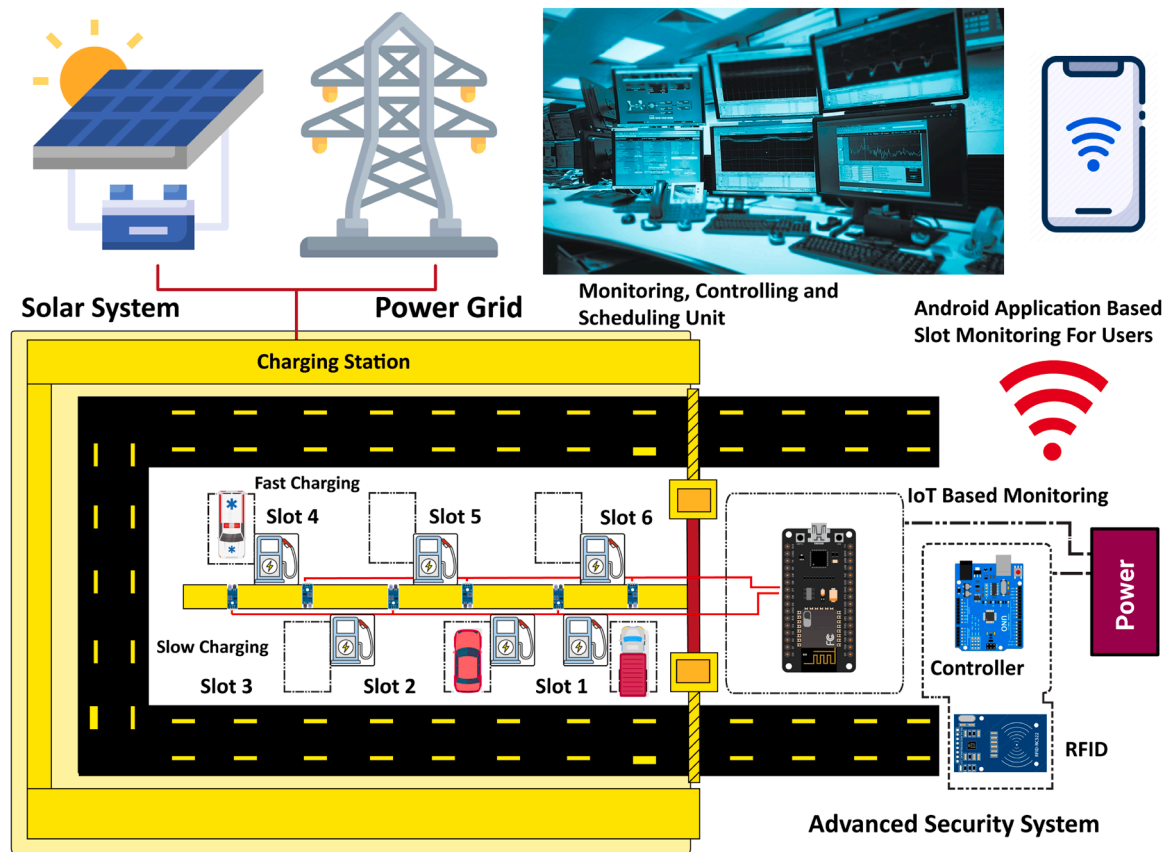


Fig. 1. Graphical abstract – revolutionizing charging stations with REI, IoT-based monitoring of charging slots for EVs scheduling, and RFID-based security system.

successfully overcome the intermittent nature of PV-based power generation.

- **Optimal Power Sharing and Energy Management:** The solution presents an approach to achieve optimal power sharing and efficient energy management at the charging station. The IoT-driven proposed system allows real-time monitoring and intelligent scheduling of EV charging sessions which leads to maximum utilization of RERs and optimal energy allocation.
- **Low-Cost Energy Generation:** The PVPPSCS introduces a solution for cost-effective energy generation for EVs. With the help of maximum utilization of solar energy, the charging station significantly reduces reliance on the grid which shows significant energy cost savings. Also, the integration of a radio frequency identification (RFID)-based security system minimizes labor costs which makes it highly cost-effective and economically viable.
- **Effective Scheduling of EV Charging:** The proposed research introduces a real-time monitoring of charging slots at a charging station system that revolutionizes the scheduling of EV charging activities. It provides real-time information about the availability of the charging slot which can save user time.
- **Advanced Security System:** This research introduces an RFID-based smart security system for charging station infrastructure. Through this RFID-based security system, unauthorized access to the charging station can be minimized and this can consequently increase the reliability of the system.
- **Reduction in Employment Costs:** In the proposed study with the integration of advanced technologies and automation the need for workers can be minimized as compared to the base case/grid-only case which does not include the integration of these technologies. This can lead to minimized operational costs and can reduce the number of workers due to automated processes with the help of IoT.

Fig. 1 represents the graphical abstract of the proposed scheme. It shows the PV-powered smart charging station optimizes energy utilization, mitigates carbon emissions, elevates the user experience, and propels the widespread adoption of EVs. With the integration of advanced technologies such as PV systems with stationary battery, IoT-based monitoring of charging slots at a charging station, and RFID-based security systems for a charging station, this research introduces a paradigm shift in the EV charging sector towards sustainable transportation.

Formation of the remaining paper

The remaining paper is structured as follows: The problem formulation of the proposed charging infrastructure is presented in Section 2. The methodology is discussed in Section 3. The results and discussion are illustrated in Section 4. Conclusions are drawn in Section 5.

Problem formulation of proposed smart charging infrastructure

This section represents the problem formulation. It involves 1) effective load management of EVs, 2) real-time monitoring of charging slots at a charging station, 3) transforming energy management with innovative EV load scheduling, and 4) a comprehensive model for RERs/PV-powered smart charging stations.

Effective load management for EVs

This section concentrates on efficient load management for EVs. It optimizes EV charging sessions by considering user preferences, charging station capacity, and the availability of energy from RERs. The objective is to minimize energy costs while maximizing the utilization of RERs.

Let T be the set of time slots representing the charging sessions, N be the set of EVs, C be the set of charging stations, P_t be the available power at time slot t from RERs and the power grid, $E_{n,t}$ be the energy demand of EV n at time slot t , $B_{n,t}$ be the battery energy level of EV n at the end of time slot t , $Cost(E_{n,t})$ be the cost function representing the energy cost for charging EV n at time slot t .

Objective and constraints: the objective is to minimize the total energy cost and maximize the utilization of RERs subject to the following constraints

1) Cost Function: The objective can be formulated as a minimization optimization problem as follows:

$$\sum (Cost(E_{n,t})) \quad (1)$$

1) Energy Demand Constraint: The energy demand constraint is represented as follows:

$$E_{n,t} \leq B_{n,t} + (E_{n,t-1} - E_{n,t}) + \sum (E_{k,t}) \quad (2)$$

for all EVs n and time slots t , where k is the set of EVs charging at the same station as EV n during time slot t .

1) Power Availability Constraint: The power availability constraint can be represented as follows:

$$E_{n,t} \leq P_t \quad (3)$$

for all EVs n and time slots t .

1) Charging Station Capacity Constraint: The charging station capacity constraint is expressed as follows:

$$\sum (E_{n,t}) \leq C \quad (4)$$

for all charging stations C and time slots t .

1) Battery Energy Level Constraint: The constraint for battery energy level is stated as follows:

$$B_{n,t} \geq B_{n,t-1} - E_{n,t} \quad (5)$$

for all EVs n and time slots t .

Real-time monitoring

This section provides a detailed description of the need for real-time monitoring of charging slots. The goal of the proposed work is to provide real-time information about the slot availability to both users and charging station operators. Binary variables are used to indicate the availability of slots and the assignment of EVs to slots which guarantees that each slot is assigned to a single EV. The representations define constraints related to the availability of charging slots and user assignments.

Let S_t be a binary variable representing the availability of a charging slot at time slot t , ($S_t = 1$ if slot available, 0 otherwise), $M_{n,t}$ be a binary variable shows the assignment of EV n to charging slot t , ($M_{n,t} = 1$ if EV n assigned to slot t , 0 otherwise).

Objective function and constraints: the objective is to provide real-time information about the availability of charging slots which subject to the following constraints

$$\text{Maximize} \sum_{t \in T} \sum_{n \in N} M_{n,t} \cdot S_t \quad (6)$$

1) Slot Availability Constraint: The slot availability constraint can be represented as follows. $S_t = 1$ if and only if:

$$\sum (M_{n,t}) \leq 1 \quad (7)$$

for all time slots t .

1) User Assignment Constraint: The user assignment constraint can

be expressed as follows.

$$\sum (M_{n,t}) = 1 \quad (8)$$

for all EVs n .

1) Slot-User Assignment Constraint: The slot-user assignment constraint can be represented as:

$$M_{n,t} \leq S_t \quad (9)$$

for all EVs n and time slots t .

Smart security system

This section provides a detailed description of designing a smart security system for the charging station. The IoT-based system guarantees authorized access and reduces the need for security and management staff. Binary variables are used to indicate authorized access and user presence. Equations show constraints related to security and authorized user access.

Let $A_{n,t}$ be a binary variable indicating authorized access of EV n at time slot t ($A_{n,t} = 1$ if authorized, 0 otherwise). $U_{n,t}$ be a binary variable indicating the user presence at the charging station for EV n at time slot t ($U_{n,t} = 1$ if present, 0 otherwise).

Objective function and constraints: the objective of the study is to design a security system that ensures authorized access which is subject to the following constraints

$$\text{Maximize} \sum_{t \in T} \sum_{n \in N} A_{n,t} \quad (10)$$

- 1) Security Constraint: The security constraint can be represented as follows. $A_{n,t} = 1$ if and only if EV n is authorized at time slot t .
- 2) User Presence Constraint: The user presence constraint is expressed as follows. $U_{n,t} = 1$ if and only if $M_{n,t} = 1$.
- 3) Authorized User Constraint: The authorized user constraint is represented as:

$$A_{n,t} \leq U_{n,t} \quad (11)$$

for all EVs n and time slots t .

Integration of RERs

This section addresses the integration of RERs, such as solar power, into the charging station infrastructure. The objective is to optimize the sharing of power between RERs, battery energy storage system (BESS), and the power grid. Equations represent power production from RERs, power sharing, and constraints related to power availability.

Let R_t be the power production from RERs at time slot t . S_t be the power-sharing between RERs, BESS, and the power grid at time slot t .

Objective function and constraints: the objective is to optimize the power-sharing, subject to the following constraints

$$\text{Optimize} \sum_{t \in T} \sum_{n \in N} R_t \cdot S_t \quad (12)$$

1) Power Sharing Constraint: The power sharing constraint is represented as follows:

$$S_t = R_t + \sum (E_{n,t}) \quad (13)$$

for all time slots t .

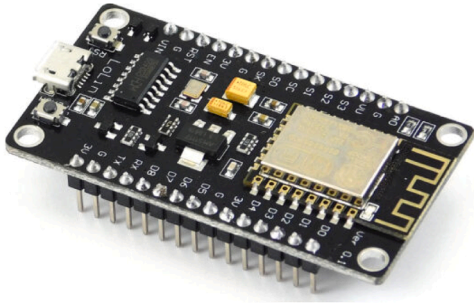


Fig. 2. The ESP8266 module – a crucial component for IoT-based monitoring in smart charging stations.

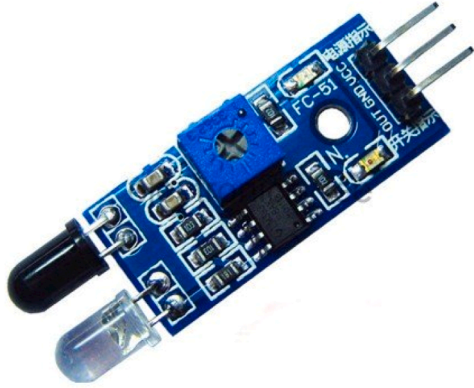


Fig. 3. Infrared sensors for precise vehicle detection in smart charging stations.

1) Power Grid Constraint: The power grid constraint is expressed below.

$$S_t \leq P_t + G_t \quad (14)$$

for all time slots t , where G_t is the available power from the grid.

Cost-effectiveness

The emphasis of this section is on achieving cost-effectiveness in the charging infrastructure. This involves maximizing the utilization of solar energy, reducing reliance on the grid, and achieving cost savings. Binary variables represent the utilization of RERs, and equations capture constraints related to cost savings and RERs utilization.

Let G_t be the power from the grid at time slot t , K be the cost savings achieved by utilizing RERs. $W_{n,t}$ be a binary variable indicating whether EV n is using RERs for charging at time slot t ($W_{n,t} = 1$ if using RERs, 0 otherwise).

Objective function and constraints: the objective is to minimize the overall cost, subject to the following constraints

$$\text{Minimize } \sum_{t \in T} (G_t - K) \quad (15)$$

1) Cost Constraint: The cost constraint is expressed as below.

$$\Sigma(\text{Cost}(E_{n,t})) + K \leq G_t \quad (16)$$

for all time slots t .

1) RERs Utilization Constraint: The constraint towards RER utilization is represented as follows.

$$E_{n,t} \leq W_{n,t} * R_t \quad (17)$$

for all EVs n and time slots t

The proposed research aims to solve the above optimization problems to achieve effective EV scheduling, real-time monitoring, a smart security system, integration of RERs, and cost-effectiveness in the EV charging infrastructure.

Methodology

This section represents the methodology. It represents 1) enhancing efficiency and cost-effectiveness with real-time monitoring of charging infrastructure, 2) energy security in smart charging station through RFID technology, 3) a smart security system, 4) integration of RERs, and 5) cost-effectiveness.

Enhancing efficiency and cost-effectiveness: real-time monitoring of charging infrastructure

Real-time monitoring of charging stations is a crucial and fundamental component of energy management. It assures the operational efficiency and cost-effectiveness of the charging station. Monitoring the available charging slots at the charging station helps the user to book the charging slots in advance. In this regard, an IoT-based strategy is used which provides real-time monitoring of available charging slots. The ESP8266 Wi-Fi module enables the continuous acquisition of data from sensors by the tracking system. This data is then sent to both the user's Android application and the main control room of the charging station. Specifically, infrared sensors are placed to keep a careful check on the charging slots and make sure that proper data is captured. The integration of these sensors helps in optimal utilization of RERs and energy management.

Using the Blynk app makes it possible to monitor the charging locations in the charging station in real time. This makes sure that the charging infrastructure is handled efficiently. Users can check on the condition of available charging places from their mobile devices through Blynk interface. This enables user to communicate with the charging station all the time and get information in real time about available charging slot. With this real-time monitoring tool, customers can make decisions. Fig. 2 depicts the ESP8266, and Fig. 3 displays the IR sensor that are components of the prototype.

Pseudo code for IoT-based charging station monitoring system

This section discusses the pseudo-code for the proposed monitoring system. Table 1 explains the line-by-line steps incorporated in the system. The IoT-based charging station monitoring system is designed to efficiently monitor the occupancy status of parking slots using infrared sensors. It provides real-time updates to the Blynk platform which is a cloud-based IoT service. The system first establishes a Wi-Fi connection by providing network credentials in lines # 1 to 7. Once connected, it establishes a connection to the Blynk platform using an authentication token and Wi-Fi credentials from lines # 8 to 11.

The infrared sensors are initialized and configured as input pins to read the status of each parking slot in lines # 12 to 14. The system then enters a main loop where it continuously monitors the sensor status and shows updates the Blynk platform whenever a change in occupancy is detected. This is made in lines # 15–33. Virtual pins on the Blynk platform indicate whether each slot is occupied or vacant. The system also ensures constant operation by periodically reinitializing the Wi-Fi connection and reconnecting to the Blynk platform. Overall, the system offers an efficient and reliable solution for monitoring parking slot availability in real-time, allowing for convenient remote monitoring and management. All the functions generated in the main loop are then called out in lines # 34 to 41.

In the PV-BSCS the security of charging stations remains a paramount concern. To restrict access exclusively to authorized customers and mitigate unauthorized usage risks an RFID-based module is installed as a viable solution. Each customer is provided with a unique RFID card which requires scanning upon arrival at the charging station. Following

Table 1
Pseudo Code for IoT-based charging station.

Algorithm IoT-based charging station monitoring system.	
1.	Initialize: Wi-Fi Connection
2.	Provide the network credentials (SSID and Password),
3.	Attempt to Connect to the Wi-Fi Network,
4.	Repeat until successfully connected to the network,
5.	Wait for a short duration (e.g., 500 milliseconds),
6.	Display a connecting status indicator,
7.	Display a successful connection message.
8.	Connect to Blynk Platform ();
9.	Provide the Blynk authentication token and Wi-fi credentials.
10.	Connect to the Blynk platform.
11.	Display a successful connection message.
12.	Initialize Sensor Pins ();
13.	Specify the pins used for the IR sensors.
14.	Configure the specified pins as input pins.
15.	Main Loop ();
16.	Enter an infinite loop.
17.	Execute the Blynk platform update function.
18.	Monitor the status of each IR sensor.
19.	For each IR sensor:
20.	Read the current status of the sensor.
21.	If the current status differs from the previous status, then
22.	Update the slot status
23.	Update the corresponding virtual pin on the Blynk platform
24.	Else
25.	Repeat the sensor monitoring process
26.	End If
27.	Update Slot Status (slot index, status)
28.	Update the virtual pin associated with the specified slot index based on the status.
29.	If the status indicates an occupied slot, then
30.	Update the virtual pin to indicate an occupied slot,
31.	Else If the status indicates a vacant slot
32.	Update virtual pin to indicate a vacant slot
33.	End If
34.	Call Initialize Wi-Fi Connection ()
35.	To establish a connection to the Wi-Fi Network.
36.	Call Connect to Blynk Platform ()
37.	To connect to the Blynk platform.
38.	Call Initialize Sensor Pins ()
39.	To initialize the IR sensor pins.
40.	Call Main Loop ()
41.	To start the main execution of the parking slot monitoring system.

Enhancing Security in Smart Charging Stations through RFID Technology.

successful verification, the RFID module grants access to the authorized customer. Particularly, autonomous integration between the entrance barrier and the RFID system ensures automatic opening upon detecting an authorized customer. On the exit side, a sensor-based system has been installed that detects vehicles and quickly opens the barrier to ensure a smooth exit process. The Arduino UNO microcontroller is utilized in this implementation which acts as a vital component for the integration and coordination of the security systems. With the help of these security measures, smart charging stations effectively safeguard their operations and maintain a controlled environment.

Pseudo code for smart security system in charging stations

This section discusses the pseudo-code for the proposed smart security system for charging stations. Table 2 explains the line-by-line steps incorporated in the system. The Setup() function initializes various components, including serial communication, SPI bus, RFID, servo motors, LEDs, buzzer, and the ultrasonic sensor as depicted in lines # 1 to 8. The initialization process ensures that all required functionalities are properly configured. The main part of the logic of the system lies within the Loop() function from lines # 9 to 25. This function operates in an infinite loop, continuously executing tasks to control access and monitor the environment.

The system starts by checking if a new RFID card is present shown in lines # 10 to 13. If a new card is detected, its UID is read, and the system verifies whether the card is authorized by comparing its UID with a list of authorized card UIDs. If the card is authorized, the system proceeds to

perform a series of authorized access actions. The system moves the RFID servo, and then after a specific duration closes it again as depicted in lines # 15 to 22. If the card is unauthorized, the system executes actions suitable for unauthorized access in lines # 23 to 24. Moreover, the system integrates an ultrasonic sensor to detect objects within a specific distance in lines # 25 to 33. The system operates in an infinite loop in line 35, ensuring continuous execution of the access control and environmental monitoring functionalities. It repeatedly calls the Loop() function to maintain a responsive and secure charging station environment.

Transforming energy management with innovative EV load scheduling

The efficient scheduling of EVs within PV-BSCS is a critical aspect that requires careful consideration. In this research study, a simulation of the charging station was developed using dedicated software. The simulation includes both fast and slow charging slots which accommodate various sizes of EVs.

One of the key objectives is to optimize the power-sharing dynamics between the PV system, BESS and the grid, to meet the load demand of the charging station. This optimization process involved analyzing the power generation throughout the year and optimal power sharing between different resources throughout the day to minimize the COE. Moreover, the SoC of the BESS was closely observed in the proposed PV-BSCS. The study also focuses on the charging and discharging patterns of the BESS, aligning them with the power generation profile of the PV

Table 2

Pseudo Code for smart security system in charging stations.

Algorithm Smart security system in charging stations.	
1.	Setup():
2.	Initialize serial communication
3.	Initialize SPI bus
4.	Initialize RFID
5.	Attach servo motors to respective pins
6.	Set pin modes for LEDs and buzzer
7.	Set the initial state for LEDs and buzzer
8.	Set pin modes for the ultrasonic sensor
9.	Loop():
10.	Check if a new RFID card is present
11.	If a new card is detected:
12.	Read the UID of the card
13.	Check if the card is authorized by comparing the UID with authorized card UIDs
14.	If the card is authorized:
15.	Perform authorized access actions:
16.	Turn on the green LED
17.	Move the RFID servo to 0 degrees
18.	Delay for a specific duration
19.	Move the RFID servo to 90 degrees
20.	Turn off the green LED
21.	Delay for a specific duration
22.	Else:
23.	Perform unauthorized access actions:
24.	End if
25.	Perform ultrasonic sensor control:
26.	Generate an ultrasonic pulse
27.	Measure the duration of the echo
28.	Calculate the distance from the duration
29.	If an object is detected within a specific distance:
30.	Move the ultrasonic servo to a specific angle
31.	Else
32.	Move the ultrasonic servo to a different angle
33.	Delay for a specific duration between readings
34.	Call Setup() to initialize the system
35.	Enter an infinite loop and repeatedly call Loop() to perform system actions.

Table 3

Cost analysis of the base case system.

Cost Type	Capital (\$)	Operational (\$)	Replacement (\$)	Total (\$)
Fast Charger	0	-762,197	0	-762,197
Slow Charger	0	-1,840,000	0	-1,840,000
Simple Tariff	0	2,610,000	0	2,610,000
Total System	0	3,397	0	3,397

Table 4

Cost analysis of the proposed case system.

Cost Type	Capital (\$)	Operational (\$)	Replacement (\$)	Total (\$)
Fast Charger	0	-762,197	0	-762,197
Slow Charger	0	-1,840,000	0	-1,840,000
PV System	180,675	1,557	0	182,232
Storage System	11,500	1,293	10,160	21,575
Converter	54,548	4,701	23,143	78,036
Simple Tariff	0	1,300,000	0	1,300,000
Total System	246,723	-1,300,000	33,303	-1,020,000

system and the demand of the charging station. The examination of these intricate dynamics provides valuable insights into optimizing the scheduling and operation of EV charging. This leads to enhanced efficiency and sustainable energy management within the charging station.

A comprehensive model for smart charging stations: EVs scheduling, real-time monitoring, and IoT-enabled security

The project includes several key components to enable efficient real-time monitoring and management of the PV-BSCS. One of the main components is the ESP8266 Wi-Fi module, which plays a crucial role in acquiring and transmitting real-time information from various sensors.

The module can continuously capture data from the installed IR sensors and provide precise monitoring of the charging slots. This ensures accurate data acquisition which ensures effective management of available slots. The acquired real-time data is transmitted to two important destinations: the Android application of the user and the central control room of the charging station.

Let:

- D_{IR} = data from IR sensors
- D_{ESP} = processed data by ESP8266 module
- T_U = data transmission rate to the user's Android application
- T_C = data transmission rate to the central control room

$$D_{IR} = \alpha \times IR \text{ intensity} + \beta \quad (18)$$

$$D_{ESP} = \int_0^t D_{IR} dt \quad (19)$$

$$T_U = T_C = \gamma \times D_{ESP} \quad (20)$$

where: α , β , γ are proportionality constants.

With the help of the ESP8266 module, this advanced system ensures that real-time information is readily accessible to both users and station operators. This allows users to conveniently monitor the status of available charging slots, track slot usage, and make decisions regarding their charging needs. Another key component that enhances the real-time monitoring capabilities is the utilization of the Blynk application. This powerful tool allows users to monitor the charging slots in real-time through their smartphones or other mobile devices. This authorizes users to stay informed and make optimized decisions about their

Table 5

Comparison of proposed study with existing literature: integration of advanced technologies, renewable energy, and security measures in EV charging infrastructure optimization.

Aspect	Integration of Advanced Technologies	Optimal Power Sharing and Energy Management	Low-Cost Energy Generation	Effective Scheduling of EV Charging
Proposed Study	PV-power smart charging station integrates advanced technologies to revolutionize EV charging infrastructure. RFID-based security system ensures charging station security while minimizing unauthorized access risks.	Achieves optimal power sharing and efficient energy management through IoT-driven real-time monitoring and scheduling.	Significantly reduces reliance on the grid through maximum utilization of solar energy, leading to substantial energy cost savings.	A real-time monitoring system optimizes EV charging scheduling by providing users with slot availability information.
[37]	Integration of EVs into China's carbon markets, optimizing charging schedules based on carbon emission prices.	Focuses on optimizing charging schedules based on carbon emission prices and historical data.	Focuses on integrating EVs into carbon markets for the economic feasibility of low-carbon transitions.	Optimizes charging schedules based on carbon emission prices, but no real-time scheduling feature is mentioned.
[38]	Utilizes historical data and deep learning methods for energy management optimization.	Utilizes historical data and deep learning methods for energy management optimization but not real-time scheduling.	Examines optimization of energy management to increase net profit, but no specific emphasis on cost savings.	Utilizes historical data and deep learning methods for energy management optimization but not real-time scheduling.
[39]	Optimizes placement of plug-in EV charging stations in distribution networks using hybrid GA-PSO approach.	Optimizes charging station placement in distribution networks to minimize power losses.	Aims to minimize power losses and voltage deviation in distribution networks with EV charging stations.	Proposes optimal charging station placement but doesn't discuss real-time scheduling features.
[40]	Utilizes clustered charging stations at commercial enterprise parking lots for reduced energy demand and cost.	Reduces energy demand and cost through clustered charging stations at commercial parking lots.	Reduces energy demand and cost through clustered charging stations, but not explicitly focused on cost savings.	Focuses on reducing energy demand and cost through clustered charging stations, without real-time scheduling.
[41]	Review of fast-charging station design emphasizing efficient speed and renewable integration.	Reviews fast-charging station design but doesn't address specific cost-saving measures.	Reviews fast-charging station design but doesn't mention real-time scheduling.	Reviews fast-charging station design but doesn't mention real-time scheduling.
[42]	Addresses optimal bidding strategy selection for EV charging in an auction market using MDQN-based approach.	Proposes MDQN-based approach for optimal bidding strategy selection in EV charging auctions.	Proposes MDQN-based approach for optimal bidding strategy selection, which may indirectly impact cost.	Proposes MDQN-based approach for optimal bidding strategy selection but doesn't address real-time scheduling.
[43]	A comprehensive review of EV battery charging methodologies and proposes advanced charging architectures.	Reviews EV battery charging methodologies and proposes advanced architectures, but not focused on cost savings.	Considers charging cost minimization in scheduling, but doesn't discuss real-time scheduling features.	Considers charging cost minimization in scheduling but doesn't mention real-time scheduling.
[44]	Framework for optimal scheduling of EV owner considering car-sharing opportunities and charging cost minimization.	Considers charging cost minimization in scheduling but doesn't discuss real-time scheduling features.	Considers electricity cost reduction in campus microgrid management, but not explicitly focused on low-cost energy generation.	Considers routing optimization and charging time allocation but doesn't mention real-time scheduling.
[45]	Cost-optimal energy management model for campus microgrids considering power generation and consumption analysis.	Considers power generation and consumption analysis for cost-optimal energy management in campus microgrids.	Discusses cost calculation based on energy-based tariff but doesn't address low-cost energy generation.	Discusses electricity cost reduction and peak power management but doesn't mention real-time scheduling.
[46]	The development and simulation of a bidirectional converter-based charging station for EVs with an IoT-based system for monitoring.	Monitors EV charging parameters but doesn't specifically discuss real-time scheduling.	Integrates solar PV and VRFB energy storage for efficient charging in high-temperature environments, which may impact cost.	Integrates solar PV and VRFB for efficient charging but doesn't mention real-time scheduling.
[47]	Presents solar-battery storage integrated switchable glazing topology for efficient EV charging in high-temperature environments.	Integrates solar PV and VRFB for efficient charging but doesn't mention real-time scheduling.	Proposes solar-battery storage integrated switchable glazing topology but doesn't specifically mention security.	Proposes solar-battery storage integrated switchable glazing topology but doesn't specifically mention security.
[48]	An optimization-driven framework using deep Maxout network and fractional feedback tree algorithm for EV charge scheduling in VANET topology.	Proposes an optimization-driven framework for EV charge scheduling in VANET topology.	Proposes an optimization-driven framework for EV charge scheduling in VANET topology but doesn't specifically mention security.	Proposes an optimization-driven framework for EV charge scheduling in VANET topology but doesn't specifically mention security.
[49]	Proposes an optimal scheduling model with uncertainty consideration for EV charging load in coordination with building systems.	Considers uncertainty for EV charging load scheduling, which may indirectly impact cost optimization.	Proposes ToU tariff rate estimation for scheduling optimization but doesn't discuss real-time scheduling.	Proposes ToU tariff rate estimation for scheduling optimization but doesn't discuss real-time scheduling.
[50]	A temporal-spatial scheduling model for EVs in AC/DC distribution networks with ToU tariff rate estimation.	Proposes temporal-spatial scheduling model for EVs in distribution networks with ToU tariff rate estimation.	Optimizes mobile charger integration for cost reduction and power profile stabilization but doesn't mention real-time scheduling.	Optimizes mobile charger integration but doesn't mention real-time scheduling.
[51]	Development and evaluation of bi-directional V2G scheduling application optimizing EV charging/discharging.	Optimizes V2G scheduling application considering user preferences and grid operator demands.	Considers user preferences and grid operator demands for V2G scheduling but doesn't mention real-time scheduling.	Optimizes V2G scheduling application but doesn't mention real-time scheduling.
[52]	Utilizes energy management system for distribution systems using IoT framework and hybrid optimization technique.	Utilizes hybrid optimization technique for energy management but doesn't explicitly address low-cost energy generation.	Implements IoT-based DSM and control strategy for efficient energy management but doesn't specifically mention security.	Implements IoT-based DSM and control strategy but doesn't specifically mention security.
[53]	Implements IoT-based DSM and control strategy for efficient energy management in smart microgrids.	Implements DSM and control strategy for efficient energy management but doesn't discuss real-time scheduling.	Proposes IoT framework for energy management but doesn't specifically mention security.	Proposes IoT framework but doesn't specifically mention security.

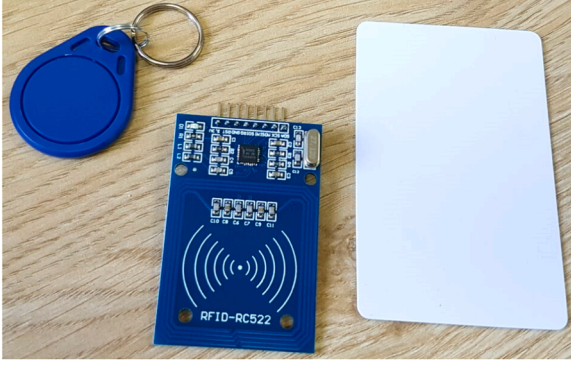


Fig. 4. Secure access with RFID module for authorized customers in smart charging stations.

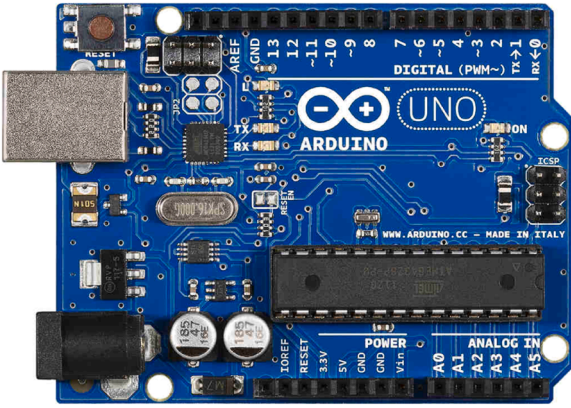


Fig. 5. Commanding control – Arduino UNO as the central controller for RFID integration in smart charging stations.

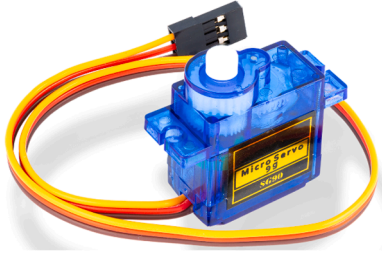


Fig. 6. Servo motor-facilitating controlled barrier operations in smart charging stations.



Fig. 7. Ultrasonic sensor enabling vehicle detection and barrier opening at the exit side in smart charging stations.

charging experience. The combination of the ESP8266 Wi-Fi module, infrared sensors, and the Blynk application creates an efficient monitoring system for the charging station. It achieves real-time communication, accurate data acquisition, and real-time monitoring of slot availability and usage. This comprehensive approach improves the performance, sustainability, and operational efficiency of the charging station.

Let:

- M_U = real-time monitoring rate on the user's mobile device
- L = latency in data display

$$M_U = \delta \times T_U \quad (21)$$

$$L = \epsilon \times (1 - M_U) \quad (22)$$

The security of PV-BSCS is another key component of supreme importance which requires comprehensive measures to restrict access exclusively to authorized customers and mitigate the risk of unauthorized usage. For that purpose, an RFID-based module is installed as a robust solution. Each customer is issued with a unique RFID card which is scanned upon their arrival at the charging station. Through a thorough verification process the RFID module grants access solely to authorized customers and permit their entry into the PV-BSCS. The integration between the entrance barrier and the RFID system ensures continuous operation and automatically opens the barrier upon detection of an authorized customer.

Let:

- R = reading from RFID card (either 1 or 0 for valid/invalid)
- A = access (either 1 or 0 for granted/denied)
- $R = \zeta \times \text{RFID card unique code}$

$$\begin{cases} 1 & R = 1 \\ 0 & R = 0 \end{cases} \quad (23)$$

Furthermore, to ensure a smooth and secure exit process, a sensor-based system has been installed at the exit point of the charging station. An ultrasonic sensor is utilized to detect the presence of vehicles. The data obtained from the ultrasonic sensor is then transmitted to the Arduino UNO microcontroller, which acts as the central control unit. The Arduino UNO microcontroller processes the information received from the ultrasonic sensor and triggers the corresponding actions. In this case, servo motors are installed to physically control the barrier at the exit. Upon detection of a vehicle by the ultrasonic sensor, the Arduino UNO sends signals to the servo motors, instructing them to rotate and open the barrier promptly. This synchronized operation ensures that authorized vehicles can exit the charging station smoothly and efficiently. With the integration of an RFID-based control system, ultrasonic sensor, and servo motors, the PV-BSCS achieves a robust and efficient security infrastructure. This advanced security setup not only safeguards the charging station's operations but also enhances the overall user experience, offering a secure and user-friendly environment for authorized customers.

Let:

- S_U = ultrasonic sensor reading
- B = barrier status,

where $B \in [0, 1]$ (0 for closed, 1 for open).

$$S_U = \eta \times \text{distance to vehicle} \quad (24)$$

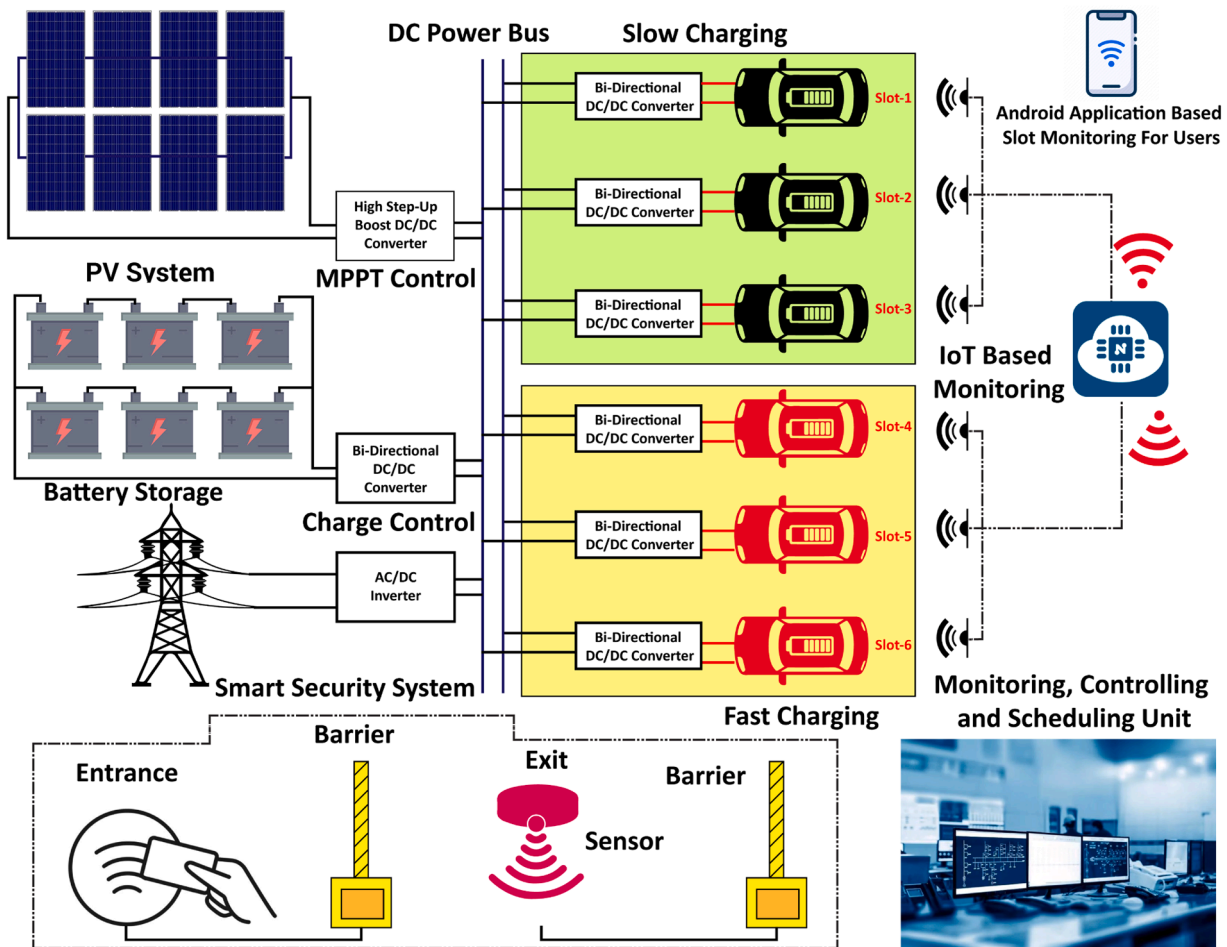


Fig. 8. System architecture of the IoT-based monitoring of charging slots and EVs scheduling solution for charging stations, showcasing the integration of IoT sensors, cloud infrastructure, EV charging stations and real-time monitoring dashboard.

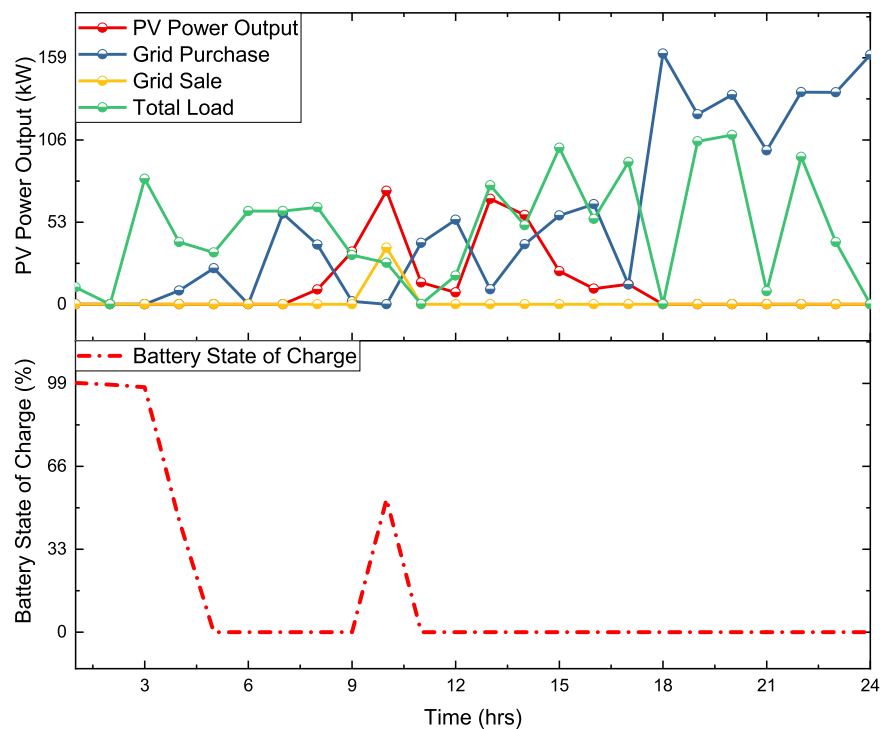


Fig. 9. Optimizing energy dynamics – January: analyzing solar PV power, battery SoC, grid purchases, and load demand.

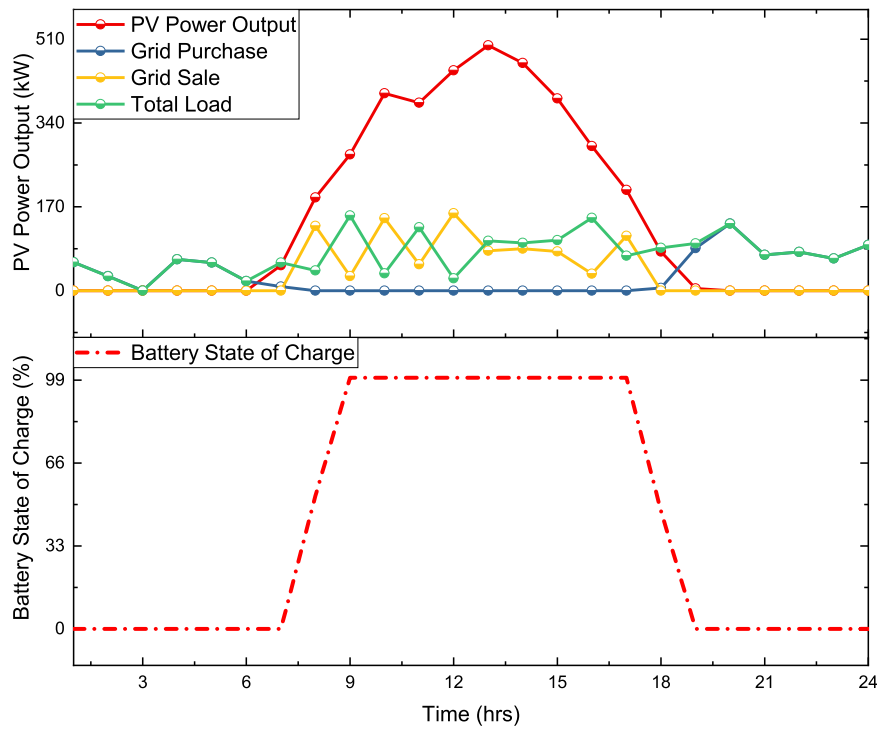


Fig. 10. Optimizing energy dynamics – April: analyzing solar PV power, battery SoC, grid purchases, and load demand patterns.

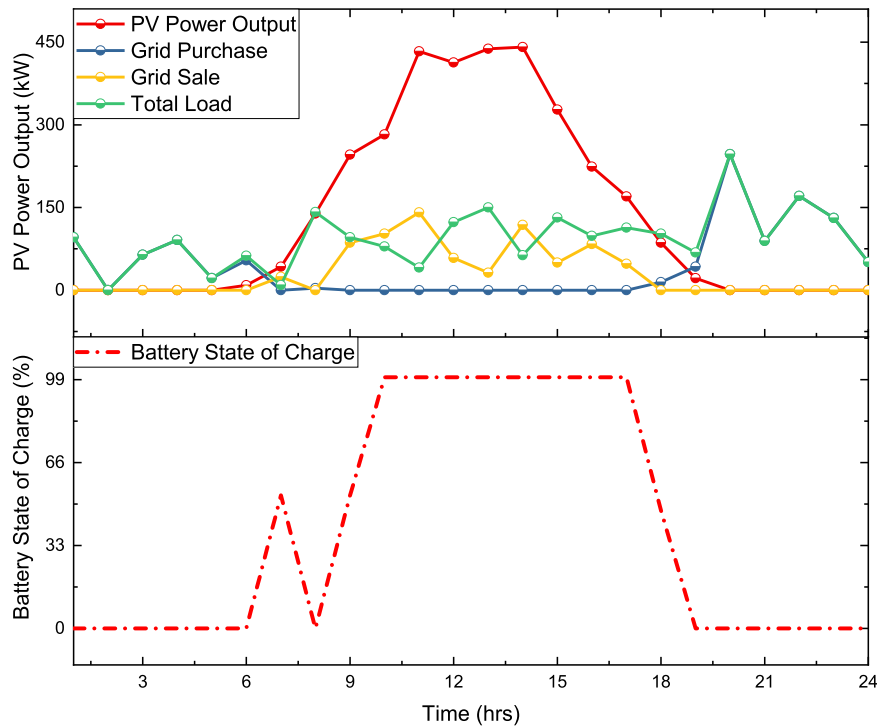


Fig. 11. Optimizing energy dynamics – July: analyzing solar PV power, battery SoC, grid purchases, and load demand patterns.

$$B(t) = \int_0^t S_0 dt \times \theta \quad (25)$$

where θ is a constant determining the barrier's response time.

The efficient scheduling of EVs within charging stations is a complex process that requires careful planning and management. In this research study, a comprehensive simulation of the charging station is done to

analyze the annualized energy production pattern of RERs through specialized software. The EVs are scheduled in a way that the maximum number of EVs can get charged by the RERs. Moreover, the BESS also provides backup assistance in combination with the grid. The optimal power sharing between the PV system, BESS and grid is achieved to meet the charging demand. The main objective was to analyze the power generation patterns throughout the year and assess the contribution of

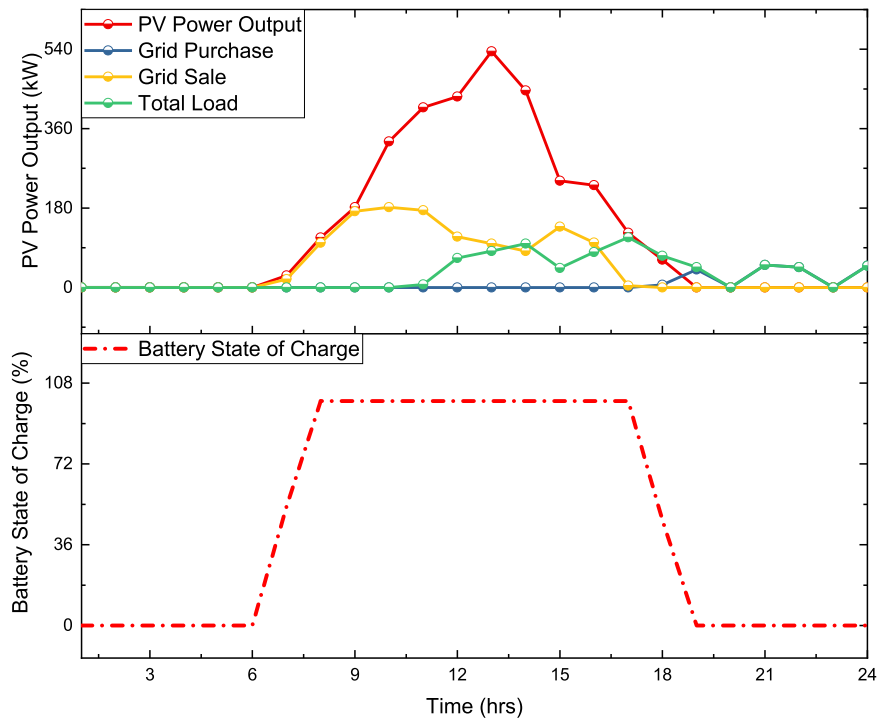


Fig. 12. Optimizing energy dynamics – October: analyzing solar PV power, battery SoC, grid purchases, and load demand patterns.

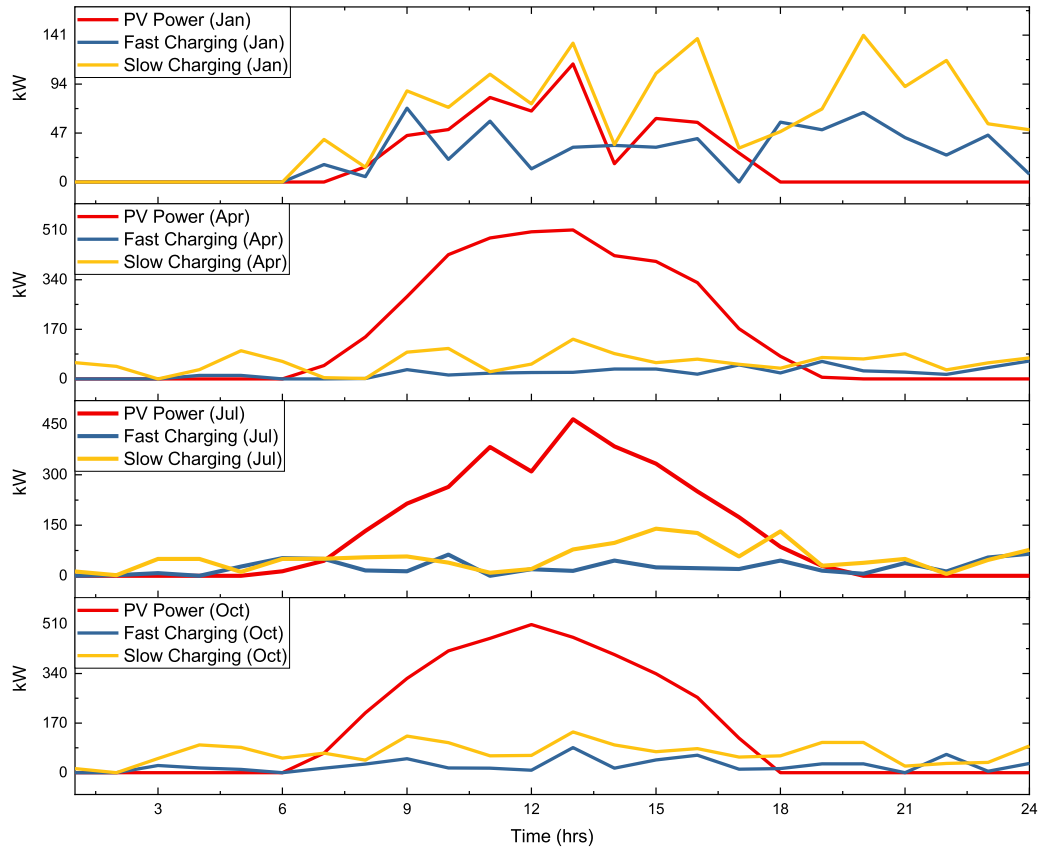


Fig. 13. Efficient load management for EV charging.

the PV system to the energy supply of the charging station. This analysis provided valuable insights into optimizing the utilization of RERs and minimizing reliance on the grid which leads to cost savings and

sustainable energy management.
Let:

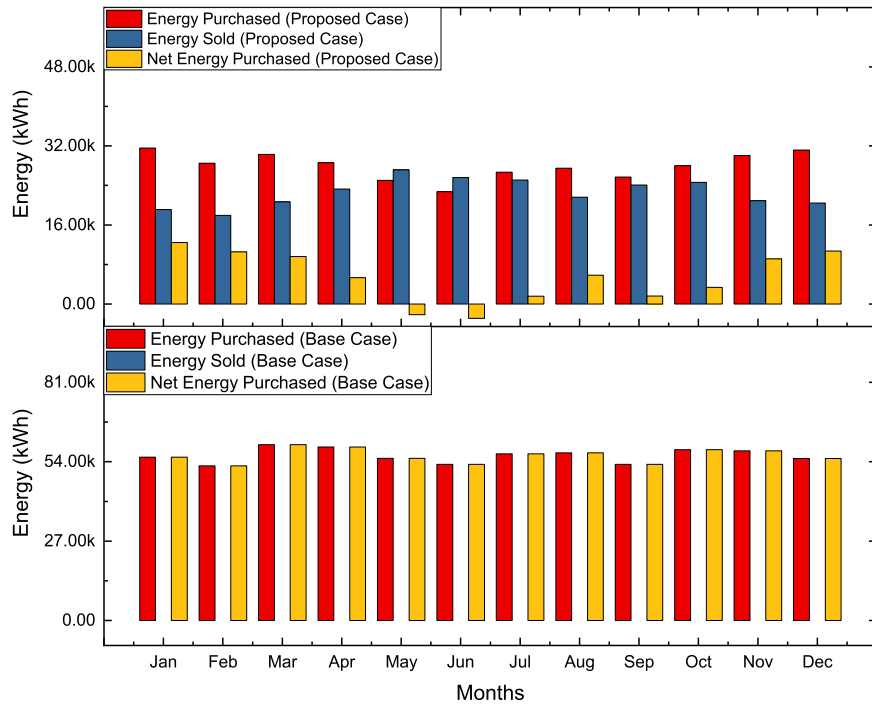


Fig. 14. Base vs. RER-integrated: grid energy comparison.

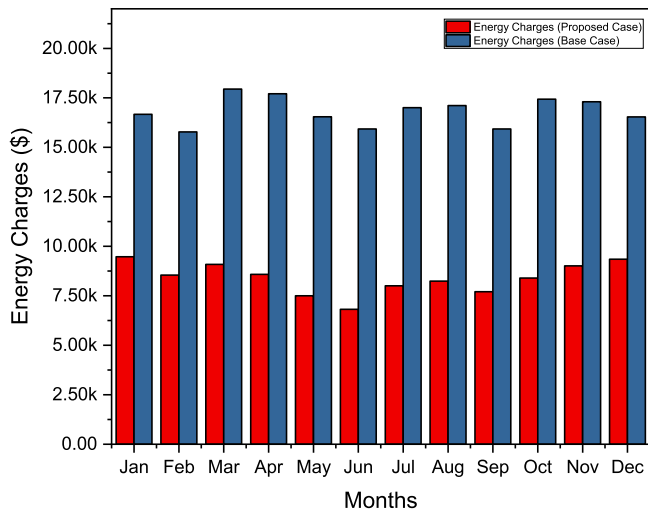


Fig. 15. Savings with RER integration: annual expenditure.

- P_{PV} = power generation by PV
- P_{BESS} = power is drawn from BESS
- P_{GRID} = power from grid
- E_{EV} = energy demand by EVs

$$P_{PV}(t) = \iota \times \text{solar intensity}(t) \quad (26)$$

$$E_{EV}(t) = \kappa \times \text{number of EVs}(t) \quad (27)$$

$$P_{BESS}(t) = \lambda \times (E_{EV}(t) - P_{PV}(t)) \quad (28)$$

where: ι , κ , λ are proportionality constants.

Furthermore, the SoC of the BESS was closely monitored and analyzed for the best possible charging/discharging of BESS. The BESS plays a crucial role in balancing the energy demand and supply within

the charging station. The charging and discharging patterns of the BESS are aligned with the power generation profile of the RERs to optimize the utilization of stored energy and minimize wastage. Through a comprehensive examination of these intricate dynamics, the study aimed to develop efficient scheduling strategies for EV charging. With the examination of the power generation from the PV system, the grid availability, and the state of charge of the BESS, the charging slots are allocated optimally. This approach would ensure that EVs are charged at the time to take advantage of RERs and reduce strain on the grid. Finally, the results of this study achieve significant scheduling and operation of EV charging within the charging station which results in improved efficiency, reduced costs, and more sustainable energy management practices.

$$E_{TOTAL}(t) = \mu \times (P_{PV}(t) + P_{BESS}(t) + P_{GRID}(t)) \quad (29)$$

$$SoC_{BESS}(t) = \int_0^t P_{BESS} dt \quad (30)$$

where μ is another proportionality constant.

Results and discussions

This section provides a complete overview of the outcomes concerning energy economics and cost estimation. power allocation between different resources and energy estimation. Furthermore, it delves into the financial and environmental ramifications associated with the implementation of the system, as well as the strategic planning of energy distribution throughout the day and year.

From grid-only to a hybrid future: case study

In this proposed study, we analyze two distinct cases for a charging station located in Islamabad, Pakistan (33°41.1'N, 73°2.9'E). The first case, termed the base case, relies solely on grid power to meet the energy demands of both slow (with output of 50 kW) and fast charging units (with output of 150 kW). The second case integrates RERs through the installation of a 602 kW PV system, complemented by a 182 kW

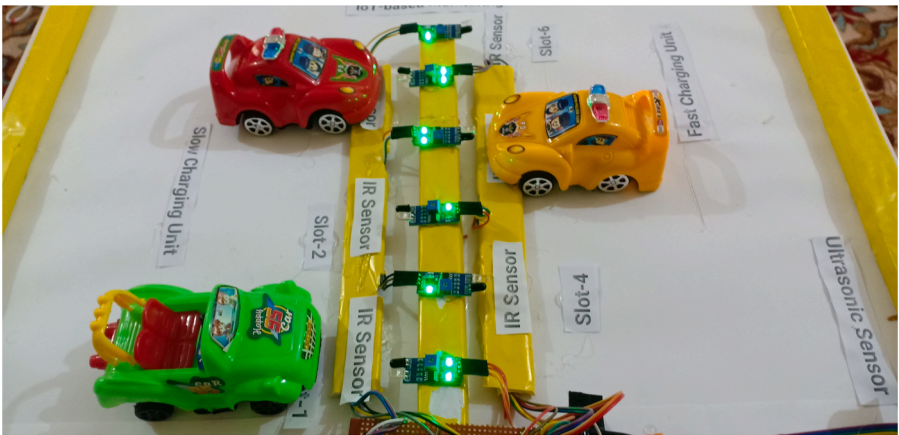


Fig. 16. Sensor-integrated IoT monitoring for charging stations.

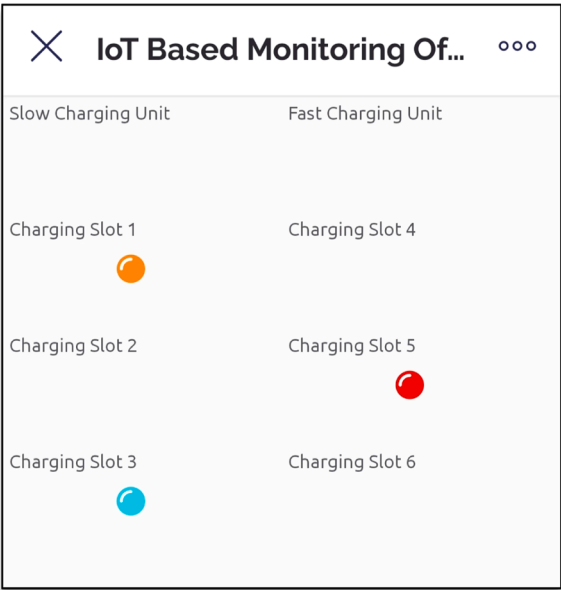


Fig. 17. Graphical user interface dashboard: IoT-based monitoring of charging stations slots.

converter and a BESS. To evaluate the potential energy generation from the PV system, we examined solar irradiance data for the location and utilized the HOMER Grid® simulation tool to model the charging station’s performance. In our analysis, we considered a total of 14,629 charging sessions annually. Specifically, the fast charger is projected to facilitate 7,333 sessions while the slow charger is expected to handle 7,296 sessions. In the base case a procurement of 672,984 kWh of energy from the grid, out of which 672,108 kWh is directly allocated for EV charging. The cost of energy (COE) is fixed at \$0.30 per kWh, and a huge amount of polluted gases are emitted with the value of 425,326 kg per year. Conversely, in the proposed setup integrated system produces a remarkable 1,055,326 kWh of energy annually, which accounts for 75.9% of the total energy consumed by the charging station. As a result, grid reliance is reduced dramatically. Only 335,775 kWh of energy is purchased, a significant reduction from the base case’s 100%. Of this generated and obtained energy, 672,108 kWh is reserved for EV charging annually which becomes 71.2% of the total energy. In the proposed case there’s a surplus energy of 28.7% of the energy, or 270,567 kWh per year, is fed back into the grid. The financial implications of this shift are compelling. The COE drops from the base value of \$0.30 to a significantly lower \$0.130, underscoring the cost-efficiency of integrating RERs into EV charging infrastructure. This not only brings about economic savings but also fosters a greener, more sustainable charging ecosystem as CO₂ emission is reduced to 212,210 kg/yr.

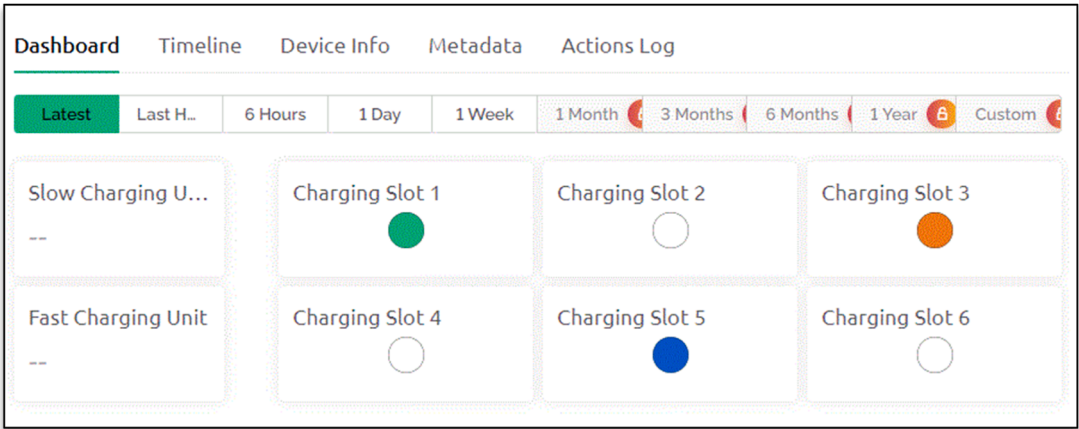


Fig. 18. Graphical user interface dashboard: Control room monitoring for enhanced charging station management.



Fig. 19. The proposed prototype for IoT-based monitoring and EVs scheduling.



Fig. 20. Developing charging station entry with RFID-based advanced access system.



Fig. 21. Charging station exit with intelligent sensor-based solutions.

Grid interaction and load balancing in solar-powered systems: monthly trends and observations

This section presents a comprehensive analysis of the estimated power production derived from RERs and the distribution of power

among different sources to meet the load demand. Additionally, the optimal charging and discharging strategy of the BESS in conjunction with various energy sources is illustrated. Power production and sharing dynamics for four different months are examined in the proposed system. In the analysis of January (as shown in Fig. 9), it is evident that PV production occurs from 07:00 a.m. to 06:00 p.m., with the highest peak at 77 kW. The BESS undergoes charging between 09:00 am and 11:00 am. Beyond 05:00 p.m., there is an increase in grid reliance due to most of the load being met by it. For April (as depicted in Fig. 10), the PV production runs from 06:00 a.m. to 07:00 p.m., reaching a peak value of 490 kW at 01:00 p.m. The BESS maintains a high SoC during the daytime, corresponding with peak PV production. The excess energy is observed to be sold back to the grid during the day.

Moving on to July (Fig. 11), the PV production reaches a maximum value of 440 kW. The BESS charges during sunshine hours and remains at peak SoC. It discharges during the night when there is no availability of RERs. In October (Fig. 12), the maximum PV production of 540 kW is achieved, and most of the load is supported by the PV system. The BESS obtains power from the PV system and discharges during the night. In this month, grid reliance is notably lower compared to January. Collectively, these observations over these four different months underscore the viability of the PV system when integrated with BESS. This combination effectively caters to EV load requirements and contributes to a reduction in the grid's burden.

Efficient load management strategies for EV charging: leveraging peak and off-peak periods to optimize resource allocation

In the presented case study, the EV load is optimized to maximize benefits from the PV production. As indicated in Fig. 13, January sees reduced PV production, resulting in the EV load being met through a combination of PV and grid sources. By April, both fast and slow charging stations are primarily powered by the PV system during the daytime. Additionally, after addressing this demand, the PV system still produces surplus energy. In July, there was a slight dip in PV production; however, it remains adequate to fulfill the EV load requirement. October also registers robust PV production, adequately meeting the demand for EVs. A breakdown of the charging dynamics reveals that the fast-charging unit achieved 7,333 charging sessions annually, consuming a

total of 196,531 kWh of energy. This led to an average energy consumption of 26.8 kWh per session. On the other hand, the slow charging unit attained 7,296 sessions over the year, using up 475,578 kWh of energy, with an average draw of 65.2 kWh per session. Cumulatively, both fast and slow charging units account for 14,629 charging sessions annually, amounting to a total energy consumption of 672,108 kWh.

IoT-enabled slot monitoring for efficient energy utilization

In the base case of the study, the charging demand of the EVs is exclusively met by grid supply, resulting in the system incurring only operational costs. The fast charger has an operational cost of -\$762,197, while the slow charger incurs a cost of -\$1,840,000 as depicted in Table 3. It's essential to note that both these values are negative, indicating the generation of equivalent revenue from both these charging units through EV charging. The energy purchased from the grid by the charging station amounts to \$2,610,000. With a total system cost of \$3,397, it becomes evident that the revenue generated by the charging station is less than the energy acquisition cost from the grid.

In the proposed case, which integrates the PV system with BESS and grid supply, there's a marked improvement in revenue generation compared to the base case. The initial investment or capital cost for the establishment of this RER-based system is approximately \$246,723 as depicted in Table 4. Interestingly, the operational cost stands at -\$1,300,000, signifying substantial revenue generation during the charging station's operation powered by RERs. Overall, the net present cost for the system is -\$1,020,000, suggesting the system will yield a revenue of -\$1,020,000 over the project's lifespan. This translates to an annualized revenue value of \$79,098. This financial analysis underscores the potential profitability and economic feasibility of adopting renewable energy sources in combination with storage solutions.

Further insights into the energy dynamics and financial implications of utilizing RERs in conjunction with the grid can be depicted with the help of Figs. 14 and 15. In the base case scenario, the annual net energy purchased from the grid stands at 672,984 kWh, incurring an annual energy expenditure of \$201,895. Notably, there is no energy being sold back to the grid in this scenario. However, when comparing this with the proposed RER-integrated system, a marked difference in energy dynamics is evident. The energy procured from the grid drops significantly to 335,775 kWh, while the energy sold back to the grid escalates to an impressive 270,567 kWh per annum. Consequently, the net energy procured is further diminished to just 65,209 kWh annually, this highlights a reduced yearly energy expenditure of \$100,733.

These statistics show the substantial economic advantage conferred by integrating RERs into the system. Specifically, there is a notable reduction of 49.89% in energy charges, underscoring the cost-efficiency and sustainability of the RERs-enhanced system.

IoT-enabled slot monitoring for efficient energy utilization

This section of the research paper introduces an innovative IoT-based monitoring system for charging stations that provides real-time monitoring to the user and the control room of charging stations. The proposed model strategically assigns three cars to slots 1 and 3 of the slow charging unit and slot 5 of the fast charging unit as depicted in Fig. 16. For charging management, the control room is equipped with a dynamic dashboard that allows real-time monitoring of the charging slots and provide a comprehensive overview of status and availability of charging station. The IoT-based user interface as illustrated in Fig. 17 allows charging station operators to efficiently allocate power resources and streamline operations of charging slots.

Similarly, an android application-based user interface is presented through a mobile application, showcased in Fig. 18 for user convenience of charging statement management. Users can effortlessly locate available charging slots at nearby charging stations from any remote location. After monitoring through the mobile application, users can book

their desired charging slot in advance, saving valuable time and eliminating the frustration of encountering occupied stations. This novel approach enhances user satisfaction by giving the user a seamless and hassle-free experience. The proposed IoT-based system also allows the charging station authorities to effectively optimize energy utilization. By strategically scheduling EVs for charging during daylight hours when PV systems yield maximum output the proposed model ensures efficient use of energy resources. This intelligent scheduling mechanism not only reduces reliance on traditional power grids but also promotes the adoption of RERs. Consequently, the charging station operates in harmony with sustainable energy practices and contributes to a greener future. The findings of this research highlight the significance of IoT technologies in transforming the charging station landscape. The real-time monitoring, user-friendly interfaces, and energy optimization features of the proposed system offer a comprehensive solution that benefits both users and station authorities. The IoT-based monitoring approach enhances user convenience and accessibility and efficient energy management empowers sustainable practices. This proposed model (Fig. 19) works for the advancement of smart charging infrastructure and paves the way for the widespread adoption of EVs.

Elevating security standards in EV charging infrastructure

In the proposed system provides a smart and automated security system for the entrance and exit points of the charging station. This feature of the PV-BSCS showcases the remarkable potential of emerging technologies in enhancing user experiences and smooth operations. At the entrance of the charging station, an advanced RFID-based module provides the user with a continuous and secure means of authentication. With a simple trace of their card, users can gain access to the charging station. The RFID module is integrated with a microcontroller which verifies the authenticity of the card and signals the opening of the entrance barrier. This intelligent integration of RFID technology and microcontroller communication is depicted in Fig. 20. Meanwhile, the exit section also exhibits an impressive feature with the integration of an ultrasonic sensor that detects vehicles at the exit point. As the vehicle is detected, the ultrasonic sensor promptly sends a signal to the Arduino Uno which enables the servo motor to open the barrier. This continuous exit process ensures a hassle-free experience for users and eliminates waiting time. These results show the effectiveness of integrating advanced technologies into charging station security systems. With the help of RFID and ultrasonic sensors as depicted in Fig. 21, the proposed system achieves a smart and automated security system.

Comparative analysis: proposed study versus existing literature on EV charging infrastructure optimization

Table 5 provides a concise comparison between the proposed study (integrated with PV system and automation technologies and smart security) and various existing papers in the domain of EV charging infrastructure optimization. Each entry highlights the unique contributions of the proposed study, which integrates advanced technologies, renewable energy utilization, real-time monitoring, and RFID-based security. Contrasted with existing research, which often emphasizes specific optimization techniques or focuses on individual aspects of EV charging, the proposed study offers a holistic solution that addresses key gaps in the literature. This comparison highlights the novel approach proposed in this study, emphasizing its potential to revolutionize EV charging infrastructure for enhanced reliability, cost-effectiveness, and sustainability.

Conclusion

In conclusion, this research paper introduces an innovative model for PV-BSCS that integrates solar power, IoT-based monitoring, and RFID-based security systems. This innovative solution transforms EV

charging infrastructure, enhancing efficiency and user experience in sustainable transportation. The optimized power sharing and energy management system achieves exceptional utilization of RERs, reduces reliance on fossil fuels and mitigates air pollution. The proposed approach has significantly reduced energy costs, from \$0.30 to \$0.130. With the integration of RERs, 75.9% of the energy required for EVs is now supplied by the PV system, with the excess energy contributing to the grid. Moreover, this system has effectively reduced GHG emissions by 49.8% with 14,629 successful charging sessions. The surplus revenue of \$79,098 is achieved through a PV system (proposed system) that supplies 75.9% of the energy for EVs, resulting in a NPC of -\$1,020,000, compared to the base case loss of \$7,803. The study also achieved a real-time monitoring system that provides detailed information about available charging slots, eliminates wait times and promotes widespread EV adoption. The advanced security system of this research ensures the reliability and integrity of the charging infrastructure through RFID-based access control for charging station security. Collectively, this research presents a comprehensive integration of advanced technologies in a smart charging station that offers a transformative solution for EV charging infrastructure. With optimized energy utilization, reduction in carbon emissions, enhanced user experience, and widespread EV adoption, this innovative charging station provides sustainable transportation infrastructure.

This research effectively integrates advanced technologies in smart charging stations but has limitations, as its findings are specific to the geographical and climatic conditions of Islamabad, Pakistan. The economic analysis also depends on current energy prices and regulations, which can change over time. Future work should involve case studies in diverse locations, incorporate additional renewable sources like wind or biomass, and explore advanced energy storage technologies to enhance system efficiency. Additionally, examining user behavior and charging patterns could optimize operations and improve user experience (Figs. 4-8).

CRediT authorship contribution statement

Anis Ur Rehman: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ishaq G. Muhammad Alblushi:** Supervision, Resources, Project administration, Formal analysis, Data curation, Conceptualization. **Haris M. Khalid:** Writing – review & editing, Resources, Project administration, Data curation, Conceptualization. **Zafar Said:** Writing – review & editing, Project administration, Data curation, Conceptualization. **Atif Iqbal:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization. **S.M. Muyeen:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare no competing financial interests or personal relationships that could have appeared to influence the work reported in this article.

Funding Statement

The publication of this article has no funding resources.

Additional Information

No additional information is available for this paper.

Data availability

Data will be made available on request.

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